Orthopolity: a general law of resource allocation

Capacity, physical realizations, and effective-resource inference

Research manuscript, revised 10 November 2026. Detailed derivations, complete comparisons and analysis histories are retained in the supporting material. Empirical studies include exploration, locally frozen forecasts and retrospective reanalyses; none is externally preregistered. Authorship and submission declarations require agreement before submission.

Abstract

Orthopolity states a general law of resource allocation: Nature tends to distribute resources equally among concentrations of those resources. We develop its mathematical content, establish exact realizations under specified physical conditions, and examine its predictive reach in models and data. Finite resource capacity bounds incidence; complete partitions give an exact inverse relation between count and mean resource. Allocation among coexisting classes requires additional conditions, including an independently defined comparison measure. Symmetric resource exchange supplies one mechanism, while gravity, radiating waves and thermal modes clarify which geometric, transport or statistical equality is present. Absorption, quantization and cell-cycle variation supply independently specified departures. The framework also supports inverse inference: restricted combinations of measured resource requirements produce identifiable predictions across environments. Tests show accurate allocation forecasts in an engineered system, pure-to-mixture chemical competition, approximate budget closure across separately grown algal cultures, and speech-duration gains from independently estimated contextual information. A conditional redistribution bound shows why changing resource prices alone cannot explain one biological failure. Other forecasts fail, and some natural catalogues cannot support the intended test after observation and estimator checks. These outcomes distinguish an established conditional relation, a physically explained departure and an unresolved failure. The contribution is a common account of resource incidence and allocation that makes those distinctions operational and connects them to further predictions. Establishing its general physical reach requires both successful transfers and explicit boundaries; failure is retained as information about the model and its domain.

1. The law and the scientific task

Nature tends to distribute resources equally among concentrations of those resources. This is the general-law statement of orthopolity. The article seeks to establish its physical content through exact ideal systems, realizations in known laws, mechanisms for real systems, and empirical predictions. It follows the accounts of Fabbri and Oliveira Jr. (2017) and Fabbri (2024), while distinguishing the claims that their shared intuition can support.

A box of volume B accommodates B/q equal smaller boxes of volume q when they tile it completely. At every complete level of a nested subdivision, more units mean less resource per unit and the same total resource. A mixed population presents an additional question: how much of the total is assigned to each coexisting size class? Conservation answers the first accounting

question but does not settle the second allocation question. Their connection, and the conditions needed to make it, are central to the law's application.

An equal allocation can appear unequal when expressed per a different coordinate, when the relevant resource combines several requirements, or when another process changes its distribution. Equal power through enclosing surfaces accompanies inverse-square radiative flux density. Equal classical energy per mode accompanies a spectrum increasing with mode density. These examples require explicit objects, resources and measures; an exponent by itself does not identify the equality. Ecological size-spectrum work provides an important precedent for making these choices physical (Sheldon et al., 1972; Gaedke, 1993; Hatton et al., 2021).

The law's scientific value also concerns departures. A standing tree experiences gravity while other forces support it; the absence of downward motion does not identify which forces act. Likewise, unequal observed allocation can coexist with an equalizing process. Establishing that explanation requires independent information about the processes and a prediction of their combined effect. A fitted residual called a constraint supplies neither. Some departures will instead expose a failed resource choice, an inapplicable mechanism or a limitation of the allocation relation in the claimed domain.

The paper develops capacity and allocation, restricted effective-resource inference, and comparisons retaining successes and failures. Additivity, reversible dynamics, equipartition and statistical inference are established foundations; the contribution concerns their physical organization, transferable predictions and limits. Supporting material retains proofs, complete comparisons and source qualifications.

2. Capacity, allocation and inference

2.1 An exact resource-incidence core

Let ν be a nonnegative additive resource measure on a system Ω, with finite budget $B = \nu(\Omega) > 0$. For disjoint units A_i with requirements $q_i = \nu(A_i)$, finite additivity gives

$$N_{\geq t} = \#\{i : q_i \geq t\} \leq B/t, \qquad t > 0.$$

Indeed, $tN_{\geq t} \leq \sum_{q_i \geq t} q_i \leq B$. This is a capacity bound; attaining it requires feasibility and use of the relevant budget.

For a nonempty finite description at level j, define its count N_j, mean resource $\bar{q}_j$, and multiplicity $m_j(x) = \sum_i \mathbf{1}_{A_{ji}}(x)$. Integration gives

$$N_j \bar{q}_j = \int_\Omega m_j \, d\nu = B\gamma_j, \qquad \gamma_j = \frac{1}{B} \int_\Omega m_j \, d\nu.$$

Every complete partition has $\gamma_j = 1$ and therefore $N_j \bar{q}_j = B$. The inverse count–mean-resource relation holds exactly even for unequal units and nested levels. Incomplete disjoint coverage has $\gamma_j < 1$; overlap can give $\gamma_j > 1$. Independently measured coverage consequently predicts a correction to the inverse relation.

A complete level accounts for the whole resource once. Several levels describe that same budget repeatedly; their totals are not independent allocations. For coexisting disjoint classes, fractions f_j instead give $N_j \bar{q}_j = B f_j$ with $\sum_j f_j \leq 1$. Equal class allocations require equal fractions, which is an additional physical statement. These elementary results establish the exact core and identify where an allocation law adds content (supporting derivation).

2.2 The allocation relation and its measure

Let k denote size on a declared finite domain and let μ be the reference measure of the classes being compared. If $n_\mu = dN/d\mu$ and $\bar{q}(k) = E[q \mid k]$ is the arithmetic mean additive resource per object, then resource density obeys

$$\mathcal{O}_\mu(k) = \frac{dR}{d\mu} = \bar{q}(k) n_\mu(k). \tag{1}$$

Neutral orthopolic allocation relative to μ has

$$\mathcal{O}_\mu(k) = C > 0, \qquad n_\mu(k) = \frac{C}{\bar{q}(k)}. \tag{2}$$

This concerns an allocation density, an expectation or a time average as specified by the system. Those interpretations are not interchangeable. Resource-weighted sampling gives a uniform size distribution under (2), while sampling objects uniformly gives an inverse-cost distribution. Stocks, accumulated requirements and throughputs must also be distinguished before applying the relation.

The reference measure is physical input. If $\bar{q}(k) \propto k^d$, equal resource per $d \ln k$ gives the **log-orthopolic** count density $dN/dk \propto k^{-(d+1)}$. Equal resource per dk instead gives the **linear-orthopolic** density $dN/dk \propto k^{-d}$. A change of coordinate carries a Jacobian and need not preserve a visually flat plot. Choosing the measure after seeing an abundance profile can manufacture equality, so an empirical claim must justify that choice independently.

One sufficient allocation condition is exchangeability of resource shares among K fixed equally weighted classes, with fixed total R. Symmetry gives $E[R_i] = R/K$ and, for fixed positive cost q_i, $E[N_i] = R/(Kq_i)$. Conservation alone does not imply this symmetry. Nor does equal expectation imply equal snapshots: placing the entire resource in one randomly chosen class has symmetric expectations and unequal realizations. The observation model must represent this distinction.

2.3 Effective resources, constraints and transferable predictions

Objects can require several resources. With independently fixed reference quantities s_ℓ, write

$$Q_i = Q_0 F(q_{i1}/s_1, \ldots, q_{im}/s_m; z_i, \theta), \qquad \bar{q}_{\text{eff}}(k) = E[Q_i \mid k].$$

The total $\sum_i Q_i$ is additive over the declared object partition, but a nonlinear F need not preserve that total when objects are split or merged. Proportional subdivision requires degree-one homogeneity for invariance. If all nonnegative constituent vectors and their subdivisions are admissible, invariance under every heterogeneous subdivision forces a finite nonnegative F, with $F(0) = 0$, to be a nonnegative linear form. Nonlinear effective resources therefore need a specified physical partition or restricted composition domain. With heterogeneous objects, averaging follows composition: $E[F(X, Y)]$ generally differs from $F(E[X], E[Y])$.

Several constituent budgets also remain several constraints unless a physical model justifies reducing them. Feasibility can require $\sum_i N_i q_{\ell i} \le B_\ell$ for each constituent. A specified optimization problem may provide scarcity-weighted effective costs, while a separately supported identical population has capacity $\min_\ell B_\ell/q_\ell$. Neither scalarization nor capacity alone fixes allocation among competing types (Boyd and Vandenberghe, 2004).

Write a constrained allocation as

$$\bar{q}_{\text{eff}}(k) n_\mu(k) = Ca(k), \qquad a(k) > 0. \tag{3}$$

A mechanism or independent measurement must determine a for a forward prediction. Inversion identifies $\bar{q}_{\mathrm{eff}}/a = C/n_\mu$, up to scale. Thus abundance contains information about an effective resource, but cannot by itself separate that resource from an unrestricted constraint. Multiplying both by the same positive function leaves the prediction unchanged.

Restrictions shared across environments make the inverse use predictive. Consider the composition $Q = Q_0 \prod_{\ell=1}^{m} X_\ell^{\theta_\ell}$. Suppose independently measured constituent requirements scale as $X_{s\ell} \propto k^{D_{s\ell}}$, and a known allocation factor is $a_s \propto k^{-v_s}$. Under logarithmic allocation, a count density $k^{-(1+\alpha_s)}$ implies

$$b = D\theta, \qquad b_s = \alpha_s - v_s. \tag{4}$$

This requires deterministic class requirements, or size-independent residual moments after combining heterogeneous constituents. For unrestricted coefficients, full column rank identifies the shared composition; otherwise only some combinations may be identified. A new constituent row has a unique forecast when it lies in the row space of the calibration design. Every left-null vector gives a parameter-free consequence $\ell^\top b = 0$.

For example, calibration rows $(1, 1)$ and $(1, 2)$ identify a two-resource composition. A new row $(2, 1)$ then predicts $b_3 = 3b_1 - b_2$ before the third abundance slope is measured. Uncertainty and near-collinear constituent scalings determine whether that forecast is useful. The identification study verifies these standard inverse-problem calculations synthetically; it does not count constructed data as natural evidence. Inverse optimization and inverse statistical mechanics provide related precedents (Keshavarz et al., 2011; Habeck, 2014). Here the scientific content lies in the specified resource family and its transferred physical predictions.

3. Ideal mechanisms and physical realizations

3.1 Equalization, a constraint and its removal

Consider fixed classes with positive reference widths w_i, fixed costs q_i, and resource stocks R_i totaling B. Symmetric conductances $g_{ij} = g_{ji} \geq 0$ on a connected graph give the exchange current

$$J_{ij}^0 = g_{ij} \left(\frac{R_i}{w_i} - \frac{R_j}{w_j} \right).$$

The resulting dynamics conserve resource and relax to $R_i^0 = B w_i / \sum_j w_j$, hence $N_i^0 = R_i^0/q_i$. This mechanism selects equal resource per the declared measure. It establishes the neutral relation within its specified model class.

A fixed class preference V_i changes the transition rate from i to j to $(g_{ij}/w_i) \exp[\max(V_i - V_j, 0)]$. Detailed balance then gives

$$R_i^V = \frac{B w_i e^{-V_i}}{\sum_j w_j e^{-V_j}}.$$

The added rates balance the neutral current at constrained equilibrium. Removing the preference preserves the current state and predicts relaxation toward the neutral allocation. The generator's spectral gap supplies a convergence bound; disconnected components equalize only within their separately conserved budgets. These are established reversible-chain tools applied to resource classes (Levin and Peres, 2017).

Whole-object assembly gives asymptotic inverse-cost mean counts under symmetric saturated channels, with finite-budget corrections; mass-action kinetics on the same conserved budget

instead favor larger objects in the worked case (supporting S3.10). Hidden link storage and nonproportional storage curves can also alter relaxation or intervention outcomes despite matched baseline measurements (supporting S5.19 and S3.7).

Passive RC networks cannot have positive imaginary impedance at their driven port. An independent forecast including magnetic storage gives 12.2% curve error, below a 39.4% lower bound for any passive RC network. Physical capacitance weights define this one-circuit test; the extension preserves energy accounting (supporting S5.20), without uniquely establishing the general law.

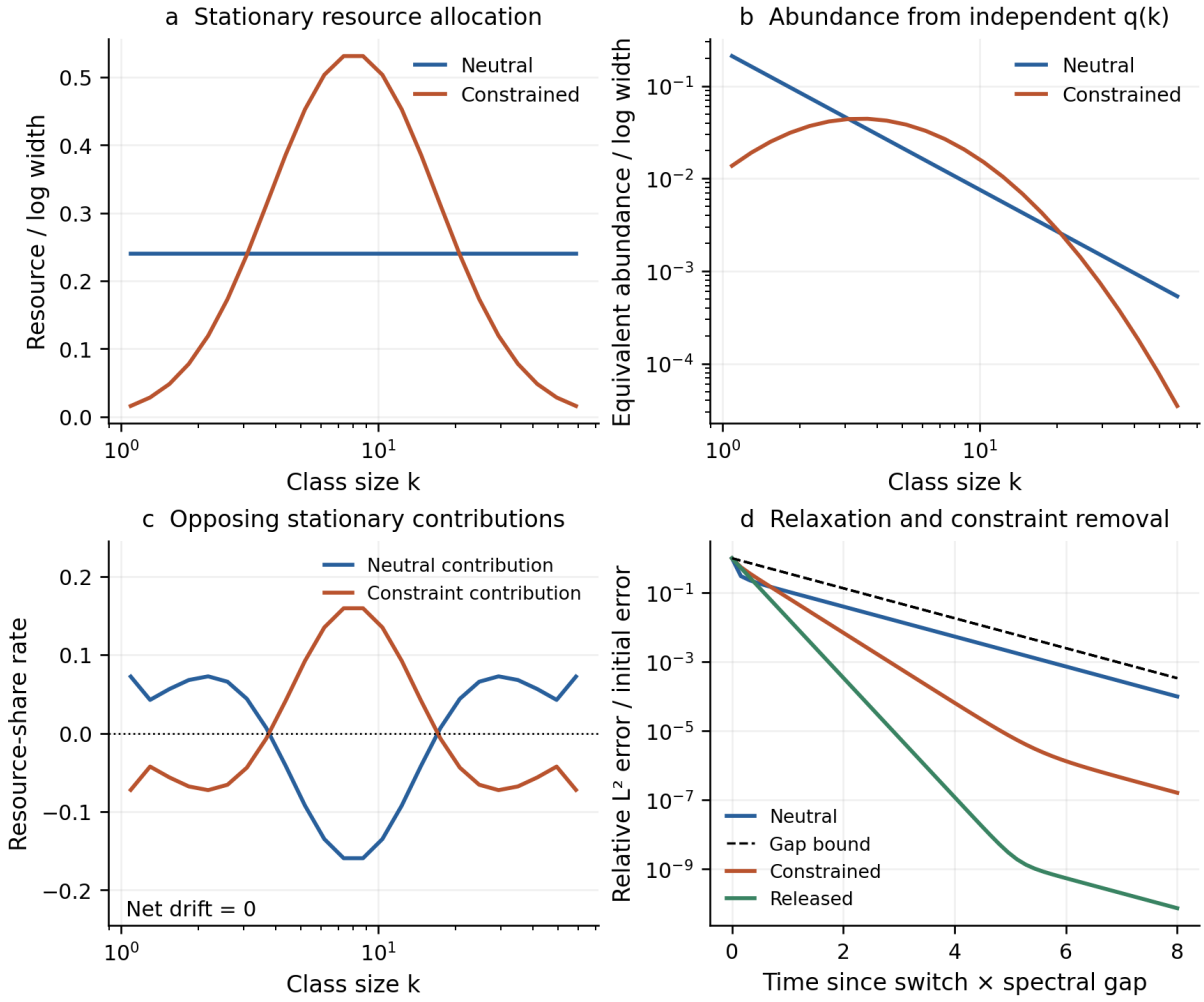


Figure 1. Resource exchange on fixed logarithmic size classes. The model predicts neutral allocation, a nonuniform constrained equilibrium and recovery after release. The detailed model, spectral bounds and finite-packet simulations are in the supporting material and model report.

3.2 Gravity and radiating waves: what geometry preserves

For a Newtonian gravitational field $\mathbf{g}$, the Poisson equation (Chao, 2014) gives the integral relation $\oint_S \mathbf{g} \cdot \mathbf{n}\, dA = -4\pi G M_{\text{enc}}$. Every enclosing surface outside a fixed isolated source has the same inward field flux. Spherical symmetry additionally gives $|\mathbf{g}| = GM/r^2$: the conserved enclosed-source quantity is distributed over an area growing as r^2. Inward flux per solid angle is $d\Psi/d\Omega = GM$, an explicit angular equality. This is a geometric realization with specified symmetry. The field flux is not an energy throughput, and nested surfaces are not independent mass or energy shares. Distributed sources change $M_{\text{enc}}(r)$; nonspherical sources need not give

uniform local fields. The gravity derivation states the resource interpretation, energy-accounting limits and departures.

For radiation, Poynting's balance gives a physical energy throughput. In steady transparent transport outside a source, the enclosing-surface power is $P(r) = L$; isotropic emission gives flux density $L/(4\pi r^2)$. With outgoing radial propagation at speed c, disjoint shell stocks instead satisfy $dU = (L/c)dr$. Equal radial thicknesses hold equal energy. Equal logarithmic radial intervals have $dU/d\ln r = Lr/c$, so they do not have equal stocks. The geometry identifies which comparison realizes equality.

An independently specified absorbing medium gives

$$P(r) = L \exp\left[-\int_{r_0}^{r} \kappa(s)\, ds\right].$$

It predicts the departure from constant radiative throughput and accounts for energy deposited in matter. If absorption is removed while the source continues, the response travels outward and the transparent profile returns by $(r - r_0)/c$. The transport derivation gives the full space-time response and its assumptions (Poynting, 1884; Haus and Melcher, 1989; Condon and Ransom, 2016). Neither the gravitational nor radiative inverse-square law alone determines the abundance of objects at different sizes.

3.3 Thermal modes: a derived limitation of neutral allocation

A finite set of classical harmonic modes in canonical equilibrium has mean energy $k_B T$ per mode. Counting modes is the independently defined measure; instantaneous mode energies can fluctuate. For electromagnetic excitation energy above the ground state, quantization gives instead

$$\bar{\epsilon}_\nu = \frac{h\nu}{e^{h\nu/(k_B T)} - 1} = k_B T \frac{x}{e^x - 1}, \qquad x = \frac{h\nu}{k_B T}.$$

The classical plateau is approached for $x \ll 1$ and suppressed for $x \gg 1$. The departure follows from the allowed states before a spectrum is observed. Mode density $8\pi\nu^2/c^3$ then gives Planck's spectrum. Equal energy per mode is therefore different from equal energy per frequency interval (Planck, 1901; Tong, n.d.).

Our FIRAS calculation converts 43 microwave-background channels to excitation energy per mode. Their $x = 1.199$–11.262 range illustrates quantum suppression. The source already incorporates a fitted blackbody template and correlated errors (Fixsen et al., 1996); this is established observational evidence, not new independent confirmation. Supporting material retains the derivation and spectrum.

For nonequilibrium cantilevers (Fontana and Bellon, 2023), colder-condition damping calibrates a quadratic local-loss model before hotter fluctuations are evaluated. Ten of eleven mode fits are physically admissible. On their 26 targets, mean absolute error is 74.0 K versus 78.9 K for uniform local loss. Large mode errors and one negative-dissipation fit show that global damping does not identify adequate spatial energy weights. Supporting S5.13 retains the retrospective design, calibration conventions, plotting omission and all failures. Inherited equilibrium normalization precludes a new equipartition test.

3.4 Cell growth and division: a measurable constraint envelope

Let cell mass grow as $\dot{x} = gx$, with common deterministic g, conservative symmetric binary division and no death. For a balanced asynchronous population and stationary lineage of physiologically equivalent daughters, growth-fragmentation theory gives $p_L(x) = xp_P(x)/E_P[x]$

(Genthon, 2022). Birth mass B and division mass D satisfy $D \overset{d}{=} 2B$. The cycle-crossing fraction is

$$H(x) = P(B \le x < D) = F_B(x) - F_B(x/2),$$

and the population law becomes

$$p_P(x) = \frac{2H(x)}{E[B^{-1}]x^2}, \qquad \frac{x^2 p_P(x)}{E_P[x]} = \frac{H(x)}{\ln 2}.$$

The right-hand side is normalized mass allocation per logarithmic size. Where every cycle crosses an interval, $H = 1$ and allocation is log-orthopolic. Fixed birth mass b gives $2b/x^2$ on $[b, 2b)$; variable births can remove the plateau. Independent birth observations predict the envelope. Growth noise, asymmetric division or lineage selection require a different observation model (derivation).

4. What the empirical comparisons establish

4.1 Predictions and comparisons

Forecasts fix objects, measure, costs, constraints and observation models. The registry and supporting material retain inputs, comparisons and prior access. Identities on the same data provide no independent confirmation; later freezing does not erase exposure.

Evidence	**Established outcome**	**Scope or unresolved limitation**
Runtime resource competition	Calibrated costs predict allocation and a thread-count intervention	One engineered mechanism
Size-selected algal cultures	Equal-biovolume forecasts transfer across selection treatments	Separate populations; binding resource unmeasured
Language transfers	Entropy predicts speech; symbolic costs transfer across genres	Known capacity account; physical cost and neutrality unidentified
Bacterial transfers	Population means and independent burden assays inform forecasts	Variability and fitted workload correction fail
Cantilever energy transfer	Some damping-to-energy forecasts transfer	Small average gain; spatial-loss and observation failures remain
Chemical / sediment constraints	Independent inputs improve transfer forecasts	Established rivals perform similarly or better; finite-accuracy limits
Chemostat nutrient pulse	Frozen two-budget reweighting fails	Constraint response needs explanation
Flare and lake geometry	Predicted exponents excluded or equivalence unresolved	Measurement and dependence affect interpretation
Seismicity	Qualification or estimator prerequisites fail	No real b–dimension comparison established

Table 1. Different studies answer different questions. Detailed results, all rival forecasts and observation limitations are retained in supporting material.

4.2 An allocation mechanism with an intervention

In a Python runtime whose global interpreter lock serializes execution, worker threads running matrix kernels of five sizes compete for one execution resource. Independent calibration gives cost degrees 1.870 and 2.686 for two kernels. The complete mean-cost curves predict CPU and job profiles before 16 allocation trials: maximum CPU-share errors are 0.0035–0.0063 and job-share total-variation errors are 0.0021–0.0124. Replacing the largest-class thread with a second smallest-class thread produces the predicted allocation change within 0.0078. A linear-cost alternative ($q_j \propto k_j$) has job-share errors of 0.126–0.273 (intervention report).

The system supplies competition and the intervention changes the independently counted opportunities. This is stronger evidence about allocation than assigning quotas and checking their cost. It demonstrates one physical implementation of a conditional resource law; it does not establish the same mechanism in nature.

4.3 Biological closure, predicted means and failed responses

In separately regrown *Dunaliella* cultures (Malerba et al., 2018), large- and small-selected treatments differ 10.4-fold in mean cell volume but only 2.5% in geometric mean total biovolume; control capacity is 18–21% higher. Frozen equal-biovolume forecasts have held-out errors of 0.124 and 0.179 log units, outperforming fitted size and equal-count alternatives. This is approximate budget closure across separate cultures; a limiting resource and allocation among coexisting classes remain unmeasured (supporting S5.3).

The cell-cycle mechanism predicts separate *E. coli* batch-culture means more accurately than lineage sampling (Gangan and Athale, 2017): errors are 2.72% and 0.447%, but variability is overpredicted and the joint criterion fails. Rounded summaries, length as a mass proxy and unverified physiological equivalence limit the transfer; supporting S5.6 retains all comparisons.

A nutrient-pulse forecast fails in twelve held-out chemostat polycultures: frozen C:N reweighting permits at most 0.05–0.12 total variation, versus observed maxima of 0.29–0.49 over days 0–12. Direction is correct in only four vessels. No forecast improves appreciably on persistence; no vessel meets the recovery criterion (Wojcik et al., 2025; supporting S5.3).

A retrospective follow-up allows arbitrary nonnegative C:N prices, preliminary quota changes and all pooled-species mixtures. Redistribution still exceeds the bound in 17/36 comparisons across eight vessels. Under transferred quotas and fixed opportunities, price adjustment alone cannot explain the failure. Separate-culture quotas do not establish mixed-vessel physiology, and assay uncertainty is unavailable. A revised explanation must change physiology, opportunities or the specified relation (supporting S5.10).

A subsequent independent-assay pilot also fails: one eligible test culture has 34.9 million cells/mL, versus 5.55 million predicted from fixed phosphorus quota and 4.24 million from an independent X-ray quota. Measurement independence does not guarantee compatible inputs (supporting S5.11). A field audit passes all 18 primary carbon/nitrogen partial-stock checks, but incomplete group coverage and undefined allocation opportunities still prevent an allocation test (supporting S5.12).

After calibration on twelve initiation variants, a separate protein-expression assay predicts ten new-protein cell responses (Borkowski et al., 2018). Under counting measure balanced over seven gene families, fitted reciprocal workload MAE is 0.10184 versus 0.09464 for direct transfer. The fitted correction fails both accuracy and advantage criteria. Direct transfer fixes the same model's environment factor at 1; neither identifies absolute resource cost (supporting S5.16).

4.4 Information costs and language transfer

Independent written-corpus entropy predicts pause-filtered speech duration through $\widehat{D} = NSID/C$, where NS counts canonical syllables and ID measures within-word conditional information (Coupé et al., 2019). Seven Indo-European languages calibrate $C = 40.40$ bits/s; sealed forecasts predict ten languages from eight other families. Under counting measure balanced over families, languages, speakers and available texts, language log MAE is 0.0741: 52.0% below constant syllabic rate, 48.8% below unigram information, 21.5% below a fitted exponent and 27.4% below same-text duration. All frozen criteria pass. Individual errors can cancel within a language; Thai's discrepancy slightly exceeds a factor of 1.20. This transfers an established capacity account, without identifying metabolic cost or neutral word allocation (supporting S5.17).

For written word types, a symbolic cost $Q_w = L_w^{\theta_L} P_w^{\theta_P}$ under counting measure gives $p(w) \propto 1/Q_w$, with letter count L and dictionary phoneme count P. The fixed composite improves held-out genres over letter power, but lexical frequencies predict better. Cost shares remain nonuniform, and these symbols do not measure acoustic energy. Supporting S5.5 retains all comparisons, including the contextual motivation of Piantadosi et al. (2011).

4.5 Geometric predictions and observation limits

Locally registered tests compare consequences fixed before outcome fitting. Aftershock productivity gives a primary exponent 0.63 with 95% interval $[0.51, 0.71]$, excluding both rupture-area and moment predictions, 1 and 1.5. Changing to rupture-length counting windows gives 0.90, $[0.76, 0.97]$. The substantial window dependence qualifies the interpretation. Flare ribbon area gives density exponent 3.163, $[2.962, 3.457]$, excluding predictions 2 and 7/3. Both registrations call these outcomes inconclusive because their rules do not choose either candidate; exclusion of the individual predictions is retained (test ledger).

Global lake radii initially give survival exponent 1.930 and iid equivalence to 2. Replacing the global projection with physical local geometry gives 1.859, with basin-bootstrap interval $[1.785, 1.924]$. Exact 2 is excluded under that analysis, but the interval overlaps the frozen equivalence band $[1.9, 2.1]$. The original positive iid conclusion does not survive this geometry correction. Two local projections agree; topology, completeness and classification still limit a physical explanation (follow-up).

The earthquake b–hypocentre-dimension comparison stops earlier. Southern California provides six sequences passing population and resolution gates, below the required ten. Italian CLASS/HORUS provides many calibrated proxies, but a frozen magnitude-estimator check fails in 45 of 56 required synthetic scenarios, including 30 of 32 noiseless converted-magnitude cases. Conversion can leave gaps that invalidate a uniform-bin correction. Neither attempt produces a real b–dimension hypothesis comparison. These are eligibility and measurement findings, not physical evidence against either relation.

4.6 Independent constraints and transport kinetics

Pure-gas calibration predicts 45 CO2/CH4 mixture conditions on reference zeolite RM8850 (Wedler et al., 2024). Shared-capacity Langmuir gives balanced component MAE 0.247 mmol/g, versus 0.620 without competition and 0.243 for established IAST. The frozen maximum-loading tolerance passes, but only 39/90 component predictions lie within reported observation uncertainties. Species/vacancy counting measure, calibrated affinities and fugacities define the unequal allocation; capacity is effective within each temperature. This supports constraint transfer

without an advantage over established adsorption theory. Supporting S5.14 retains extrapolation and retrospective-access limits.

Separate grain assays improve a transfer from 19 narrow-channel sphere runs to nine wider-channel runs (Deal et al., 2026). Measured shape reduces balanced flux error by 42.6% versus unit shape; the fuller wall correction reduces it by 77.2% versus hydraulic radius. Yet separate old-material fits predict better, low-flux errors exceed fourfold and the accuracy tolerance is permissive. One target was exposed before freezing. Material-balanced counting measure, all alternatives and qualifications are retained in supporting S5.15.

Four earlier membrane pairs predict 56 later chamber readings without a target rate fit (Benstead et al., 2026). Mean contrast error is 4.23% of initial total, versus 54.72% for persistence, 46.22% for immediate mixing and 4.14% for empirical interpolation. All-benchmark superiority fails. A separate stock test gives p=0.309; broad optical uncertainty prevents treating non-rejection as proof of conservation. Compartment volume defines the measure. This transfers ordinary diffusion kinetics, not a unique law prediction (supporting S5.18). None of these transport studies tests neutral log-/linear-orthopolic size allocation.

5. Learning from validity, departures and failure

Normal aggregation and natural allocation offer complementary explanations. Gaussian normality can emerge from additive aggregation under central-limit conditions; we call the normalized neutral inverse-resource profile (2) a *natural (orthopolic) distribution*. Homogeneous geometric requirements yield power-law abundance under the declared log-orthopolic or linear-orthopolic measure. Both mechanisms can coexist: canonical harmonic modes have Gaussian coordinates and equal mean energies. Supporting S3.9 gives the conditions and distinguishes physical scaling relations from sampled distributions.

Uniform spatial cuts of a homogeneous rod conserve every realization's mass and give labeled fragments equal expected mass. Yet expected resource per logarithmic fragment mass is nonconstant. This analytic failure of log-orthopolic size-class equality follows before outcomes, without fitting a correction. Supporting S3.8 gives the finite-bin prediction and distinguishes this partition mechanism from exchange under class symmetry.

Departures have distinct explanations. Changing the measure or sampling alters the reported distribution; independently specified absorption, quantum levels or adsorption affinities predict physical constraints. Chemostat reweighting, bacterial variability and cantilever spatial-loss errors instead remain failed forecasts until a revised explanation succeeds on additional evidence.

Unavailable tests cannot decide between resource models, and unrestricted resource fitting supplies an identity (section 2.3). Further comparisons should change independently specified constraints so that rivals predict different allocation profiles, relaxation rates or cross-environment contrasts such as (4).

6. Contribution and conclusion

Orthopolity links resource concentration, incidence and allocation as a general law. Finite capacity and complete accounting establish its exact core; specified symmetries and dynamics realize its allocation extension. Physical laws clarify the role of the measure and independently specified constraints. Restricted inverse inference then predicts how abundance transfers between environments.

The empirical results establish conditional successes, failures and observation boundaries, but not a common neutral regime across the natural populations examined. Establishing the law's

general physical reach requires a resource explanation that predicts both where it holds and how a specified change makes it depart or fail.

Data and code availability

The repository and run registry retain sources, configurations, forecasts, algorithms and outcomes, including failures. The supporting material contains detailed derivations, complete comparisons and study reports. Source licences are in data/NOTICE.md. Published FIRAS reconstruction and bacterial figure summaries remain identified as such.

The make all target verifies inputs, runs tests and regenerates the original analyses; later studies have separate offline targets in their reports. The make paper target builds both manuscripts; registry-verify checks retained study references. The supplied 2017 manuscript was inspected privately and is not redistributed.

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